

Mechanisms of Social Status Regulation as Free-Energy Minimising Loops

Compiled for AI Safety Collaborators

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Abstract

Human status dynamics appear to be governed by a portfolio of partially independent optimisation loops, each minimising a distinct component of variational free energy. We propose a compact taxonomy—from reptilian dominance reflexes to language-mediated reputation markets—and specify, for each loop, (i) the free-energy term measured, (ii) the neuroanatomical registers and actuators, and (iii) a plausible micro-mechanism that yields a prediction-error (PE) signal with neuromodulatory precision-gating. We articulate an internal/external symmetry (interoceptive threat vs. exteroceptive perimeter defence; somatic repair vs. niche construction; etc.), derive perturbation predictions, and outline how these principles transfer to opaque-agent forensics and brain-like AI benchmarks.

1 Background

Under the free-energy principle (FEP) [5], adaptive systems minimise expected surprise by building predictive models and selecting actions that make sensory data conform to those models. Status regulation can be cast as a set of social free-energy ledgers: uncertainty about threat, coalition membership, norm adherence, prestige sources, and reputational standing. Evolutionary continuity suggests layered solutions: brainstem circuits for dyadic dominance, limbic coalition bookkeeping, and neocortical narrative broadcast [12, 6, 15].

2 Loops, ledgers, and evolutionary stratification

Table 1 orders status loops from older to newer and states, for each, the free-energy target F , registers, actuators, and neuromodulators that set PE precision.

3 How circuits *measure* the ledgers

Across loops, a common micro-architecture appears: deep pyramidal cells carry priors; superficial cells compute residuals; neuromodulators scale precision (inverse variance). Concretely:

- **Dominance threat:** BLA sends expected threat to VMHvl; MeA relays opponent cues. Subtractive interneurons in VMHvl produce a PE that recruits PAG; raphe 5-HT sets risk tolerance, testosterone gates plasticity [9, 12].
- **Coalition entropy:** TPJ/mPFC encode a sparse triadic graph prior. pSTS supplies live affiliative cues. Error neurons in TPJ fire when link likelihood deviates, with OXT increasing gain (precision) on in-group edges [13, 8].
- **Prestige:** Mirror system (IFG/pSTS) hosts a predictive motor program of the model; cerebellum/PPC compute kinematic mismatch; VTA DA encodes epistemic PE that reweights imitation [11, 14].
- **Norms:** vmPFC stores rule templates; ACC compares allowed action vs. observed; an aversive PE drives dlPFC→BG sanction. Relief upon resolution corresponds to a drop in F_{norm} and engages μ -opioid [7, 3].

Age	Loop	Free-energy term F	Registers (priors / evidence)	Actuators	Precision gate
1	Dominance-submission	$F_{\text{threat}} = \Pr(\text{injury} \mid \pi)$	BLA prior $\leftrightarrow$ VMHvl, MeA evidence	PAG columns (fight/flight), spinal patterns	5-HT, T [9, 10]
2	Display-assessment	$F_{\text{signal}} = \hat{s}_{\text{self}} - s_{\text{peer}} $	Ventral striatum template $\leftrightarrow$ auditory/visual cortex	Pattern generators (laryngeal, facial)	DA [14]
3	Coalition entropy	$H(C_{ij})$	TPJ/mPFC graph prior $\leftrightarrow$ STS cues	Grooming/approach	OXT [13, 8]
4	Prestige-biased learning	$F_{\text{ep}}^{\text{social}} = \mathbb{E} o - \hat{o}_{\text{model}} $	IFG+pSTS mirror prior $\leftrightarrow$ PPC/cerebellar mismatch	Imitation weighting, teaching	DA [11, 1]
5	Norm enforcement	$F_{\text{norm}} = \Pr(\text{rule breach})$	vmPFC/hippocampal rule prior $\leftrightarrow$ ACC/insula error	dlPFC $\rightarrow$ BG punishment policy	μ -opioid, NA [4, 7, 3]
6	Reputation broadcast	$F_{\text{rep}} = \hat{r}_i - r_i $	DMN (PCC, temp. pole) prior $\leftrightarrow$ STG language input	Narrative update, signalling	DA/ OXT [15, 6]

Table 1: Status loops from evolutionarily older (1) to newer (6). Abbreviations: BLA basolateral amygdala; VMHvl ventromedial hypothalamus (ventrolateral); MeA medial amygdala; TPJ temporoparietal junction; STS superior temporal sulcus; IFG inferior frontal gyrus; PPC posterior parietal cortex; dlPFC dorsolateral prefrontal cortex; BG basal ganglia; DMN default mode network; NA noradrenaline; OXT oxytocin; DA dopamine; 5-HT serotonin; T testosterone.

- **Reputation:** DMN maintains a semantic centrality prior; language cortex updates evidence; ventral striatum emits a value PE proportional to change in expected future cooperation [15].

4 Internal $\leftrightarrow$ external symmetry

Each interoceptive ledger has an exteroceptive twin: internal threat (amygdala/PAG) vs. external perimeter defence (SC-guided vigilance); internal somatic repair vs. external niche construction; internal norm adherence (guilt) vs. external sanction (punishment). This symmetry follows from FEP at two scales: body maintenance and niche shaping.

5 Perturbation predictions

TMS to dlPFC (norm loop)

Continuous theta-burst TMS over right dlPFC should attenuate third-party punishment without reducing ACC mismatch (N200/P300). Behaviourally, acceptance of unfair splits rises; sanction spending falls [7]. Relief signals (opioid) diminish until excitability recovers.

Serotonergic shift (dominance loop)

An acute SSRI dose (increased 5-HT) raises the submit threshold and reduces aggressive approach under status threat; the PE amplitude in VMHvl drops for identical opponent cues [10].

6 Consequences for agency and consciousness

Status loops feed into valuation-efference circuits: changes in F_{threat} and $H(C_{ij})$ directly alter discounting, exploration, and social policy selection [2]. Global broadcast and metacognition render these computations

available to report, shaping conscious feelings of pride, shame, outrage. Miscalibrated precision (e.g., high OXT precision on in-group edges) can lower social free energy locally while increasing global risk (out-group hostility), a classical alignment pitfall for artificial agents.

7 Opaque-agent forensics: low-sample discovery

Borrowing from lesion logic, we recommend four probes per discovered module: (i) topology screen (hub vs. chain), (ii) one-bit poke (rebound gain $g > 1$ indicates homeostasis), (iii) 30 s behavioural battery (rank cue, norm breach, prestige honeypot), and (iv) power snapshot. A triple-hit (structure+PE+behaviour) labels a status loop without long recordings.

8 Outlook

The status stack recapitulates vertebrate evolution while exploiting neocortical broadcast. Incorporating these ledgers into AI benchmarks should surface prestige-seeking, coalition maintenance, and norm enforcement as distinct optimisation targets, enabling targeted “cooling” or constraint of each loop.

Limitations. Citations are indicative; multiple parallel routes can implement the same PE computation. Quantitative priors (precision) are context-dependent and ethically sensitive to manipulate in humans.

References

- [1] Maciej Chudek and Joseph Henrich. Culture-gene coevolution, prestige and conformity. *Philosophical Transactions of the Royal Society B*, 366:1149–1158, 2011.
- [2] Nathaniel D. Daw et al. Cortical substrates for model-based vs. model-free learning. *Nature Neuroscience*, 8:1704–1711, 2006.
- [3] Naomi I. Eisenberger et al. Does rejection hurt? *Science*, 302:290–292, 2003.
- [4] Ernst Fehr and Simon Gächter. Altruistic punishment. *Nature*, 415:137–140, 2002.
- [5] Karl Friston. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2): 127–138, 2010.
- [6] Joseph Henrich and Francisco Gil-White. The evolution of prestige. *Evolution and Human Behavior*, 22:165–196, 2001.
- [7] Daria Knoch et al. Disruption of right dlpc decreases norm compliance. *Science*, 314:829–832, 2006.
- [8] Michael Kosfeld et al. Oxytocin increases trust. *Nature*, 435:673–676, 2005.
- [9] Dayu Lin et al. Functional identification of an aggression locus in the mouse hypothalamus. *Nature*, 470:221–226, 2011.
- [10] Michael J. Raleigh et al. Serotonergic mechanisms in vervet dominance. *Brain Research*, 559:181–190, 1991.
- [11] Giacomo Rizzolatti and Laila Craighero. The mirror-neuron system. *Annual Review of Neuroscience*, 27:169–192, 2004.
- [12] Robert Sapolsky. Social status and health in humans and other animals. *Annual Review of Anthropology*, 33:393–418, 2004.
- [13] Johanna Scholz et al. rtpj in attention and social cognition: combined fmri-tms. *Cerebral Cortex*, 26: 4282–4291, 2016.

- [14] Wolfram Schultz et al. A neural substrate of prediction and reward. *Science*, 275:1593–1599, 1997.
- [15] Caroline F. Zink et al. Know your place: neural processing of social hierarchy. *Neuron*, 58:273–283, 2008.